K-LINEAR EXACT FUNCTORS ON COHERENT MODULES OVER FINITE TYPE SCHEMES ARE FM FUNCTORS WITH KERNELS FINITE OVER Y

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original equation 0FZC: sources/stacks-chow/tags.txt, source lines 19947–19947. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZH: sources/stacks-chow/tags.txt, source lines 19952–19952. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 08DS: sources/stacks-chow/tags.txt, source lines 10647–10647. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZL: sources/stacks-chow/tags.txt, source lines 19956–19956. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZJ: sources/stacks-chow/tags.txt, source lines 19954–19954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05JV: sources/stacks-chow/tags.txt, source lines 7179–7179. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZM: sources/stacks-chow/tags.txt, source lines 19957–19957. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X, Y be finite type schemes over k with X separated. There is an equivalence of categories between*

(1) *the category of k-linear exact functors $F : Coh(\mathcal{O}_X) \to Coh(\mathcal{O}_Y)$, and*

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(2) *the category of coherent $\mathcal{O}_{X\times Y}$-modules $\mathcal{K}$ which are flat over X and have support finite over Y*

given by sending $\mathcal{K}$ to the restriction of the functor (0FZC) to $Coh(\mathcal{O}_X)$.

Proof. Let $\mathcal{K}$ be as in (2). By Lemma 0FZH the functor F given by (0FZC) is exact and k-linear. Moreover, F sends $Coh(\mathcal{O}_X)$ into $Coh(\mathcal{O}_Y)$ for example by Cohomology of Schemes, Lemma 08DS.

Let us construct the quasi-inverse to the construction. Let F be as in (1). By Lemma 0FZL we can extend F to a k-linear exact functor on the categories of quasi-coherent modules which commutes with arbitrary direct sums. By Lemma 0FZH the extension corresponds to a unique quasi-coherent module $\mathcal{K}$, flat over X, such that $R^q\mathrm{pr}_{2,*}(\mathrm{pr}_1^*\mathcal{F}\otimes_{\mathcal{O}_{X\times Y}}\mathcal{K}) = 0$ for $q > 0$ for all quasi-coherent $\mathcal{O}_X$-modules $\mathcal{F}$. Since $F(\mathcal{O}_X)$ is a coherent $\mathcal{O}_Y$-module, we conclude from Lemma 0FZJ that $\mathcal{K}$ is coherent.

For a closed point $x \in X$ denote $\mathcal{O}_x$ the skyscraper sheaf at x with value the residue field of x. We have

$$F(\mathcal{O}_x) = \mathrm{pr}_{2,*}(\mathrm{pr}_1^*\mathcal{O}_x \otimes \mathcal{K}) = (x \times Y \to Y)_*(\mathcal{K}|_{x\times Y})$$

Since $x \times Y \to Y$ is finite, we see that the pushforward along this morphism is faithful. Hence if $y \in Y$ is in the image of the support of $\mathcal{K}|_{x\times Y}$, then y is in the support of $F(\mathcal{O}_x)$.

Let $Z \subset X \times Y$ be the scheme theoretic support Z of $\mathcal{K}$, see Morphisms, Definition 05JV. We first prove that $Z \to Y$ is quasi-finite, by proving that its fibres over closed points are finite. Namely, if the fibre of $Z \to Y$ over a closed point $y \in Y$ has dimension > 0, then we can find infinitely many pairwise distinct closed points $x_1, x_2, \ldots$ in the image of $Z_y \to X$. Since we have a surjection $\mathcal{O}_X \to \bigoplus_{i=1,\ldots,n} \mathcal{O}_{x_i}$ we obtain a surjection

$$F(\mathcal{O}_X) \to \bigoplus\nolimits_{i=1,\ldots,n} F(\mathcal{O}_{x_i})$$

By what we said above, the point y is in the support of each of the coherent modules $F(\mathcal{O}_{x_i})$. Since $F(\mathcal{O}_X)$ is a coherent module, this will lead to a contradiction because the stalk of $F(\mathcal{O}_X)$ at y will be generated by $< n$ elements if n is large enough. Hence $Z \to Y$ is quasi-finite. Since $\mathrm{pr}_{2,*}\mathcal{K}$ is coherent and $R^q\mathrm{pr}_{2,*}\mathcal{K} = 0$ for $q > 0$ we conclude that $Z \to Y$ is finite by Lemma 0FZM. □